Smart Grocer Statistical Analysis

Ellie Burton · Capstone in Data Science · Spring 2026

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Project Overview

Motivation

- Grocery costs for college students
- Automate tedious efforts of checking prices online
- Identifying the best grocery plan

Research Questions

1. Is there a significant **difference in basket cost** between retailers?
2. Does retailer effect remain consistent **across product categories**?
3. Which products & retailers exhibit the highest price **volatility**?
4. Is there a **day-of-week** effect on grocery prices?
5. Can total cost be **minimized** by splitting a basket across stores?

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Project Overview

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Data at a Glance

1,332 total rows	**37** days: Feb 3 – Mar 11
12 basket items tracked	**3** retailers compared

Research Questions

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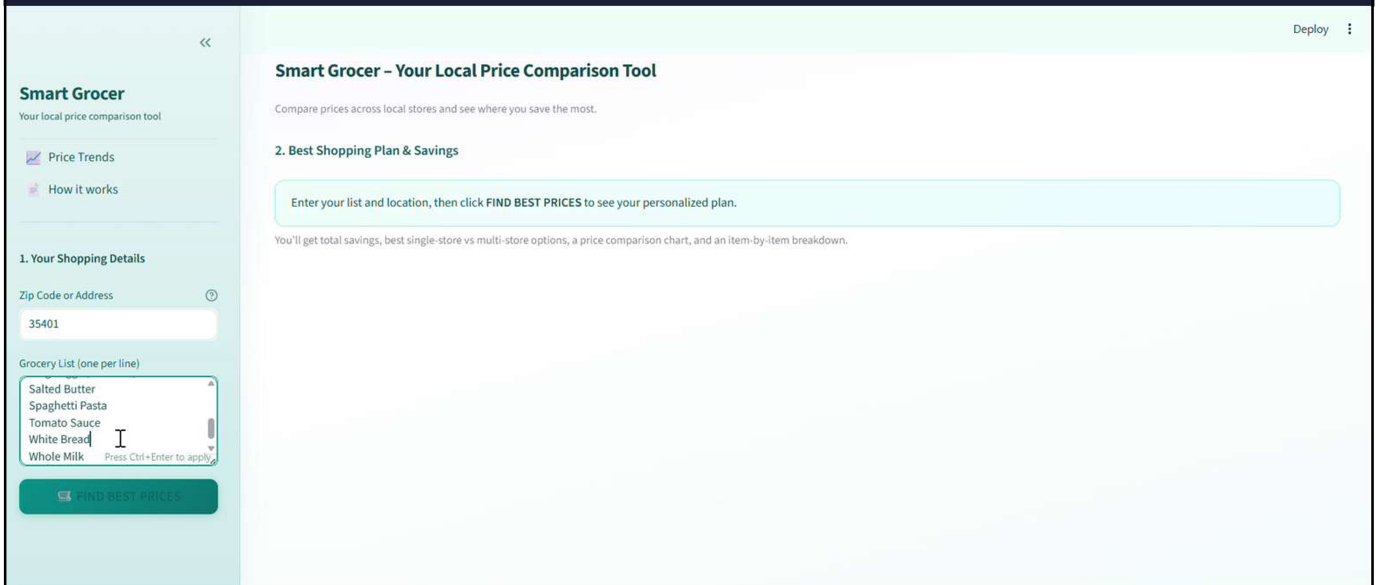
01 DATA & METHODS

Data Collection, Enrichment Pipeline & Cleaning

Demo



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Technical Application

UI

- Enter a grocery list and zip code to get real-time price comparisons across Aldi, Publix, and Walmart
- Prices scraped live using Chrome-based web scrapers
- Results displayed as a sortable comparison table to see which store is cheapest for each item at a glance

Script

- Daily basket script supports 12 staple items and can be scheduled with Windows Task Scheduler
- Data saved to CSV for longitudinal tracking via run_daily_basket.py

Streamlit

Web UI framework

Selenium

Chrome scraping

Python 3.9+

Core runtime

pandas / scipy

Data + stats

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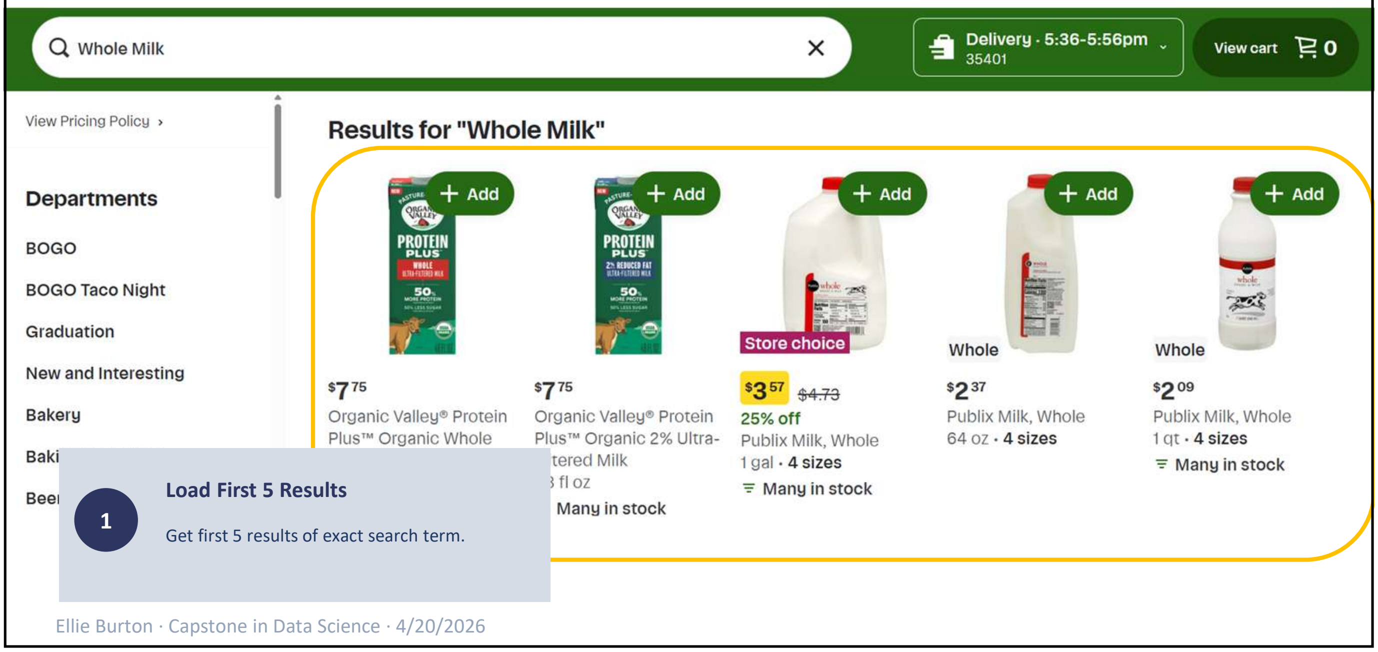
Data Enrichment Pipeline



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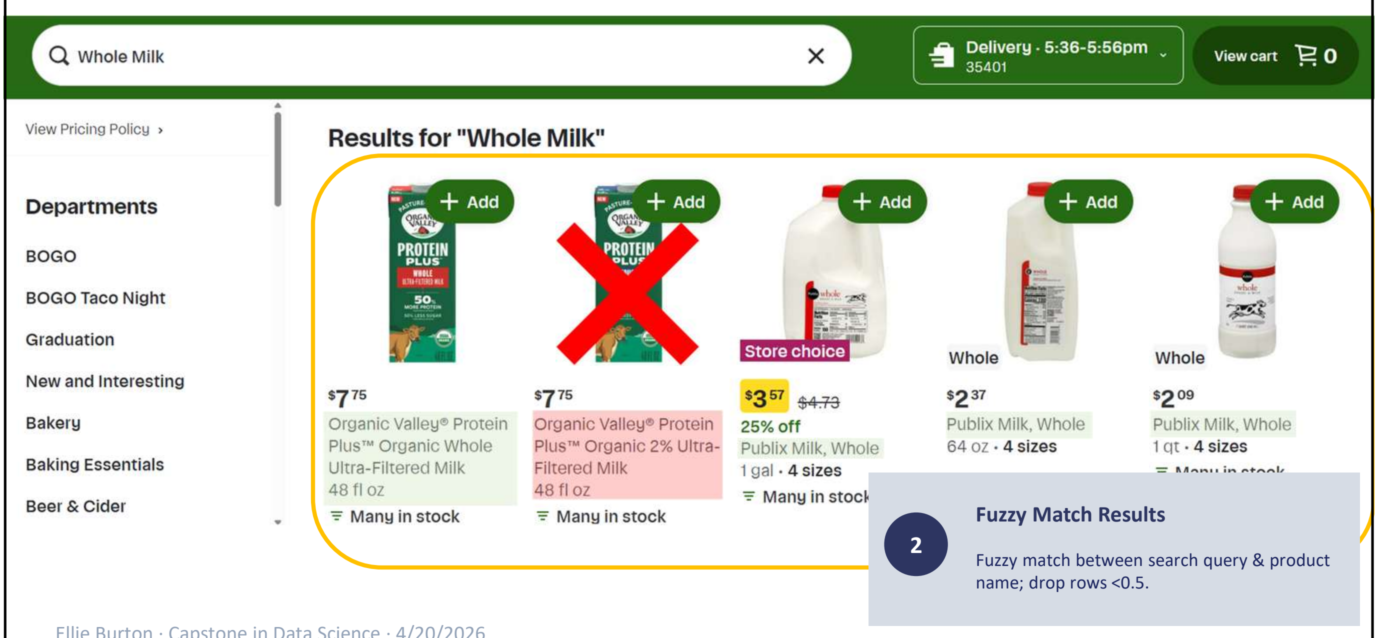
Load First 5 Results

Get first 5 results of exact search term.

Data Enrichment Pipeline



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Fuzzy Match Results

Fuzzy match between search query & product name; drop rows <0.5.

Data Enrichment Pipeline

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Store raw data

Field	Raw scraped data
search_term	Whole Milk
product_name	Publix Milk, Whole
price	$3.57 25% off
unit_size	1 gal
store	Publix

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Data Enrichment Pipeline

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Clean data

Field	Raw scraped data
search_term	Whole Milk
product_name	Publix Milk, Whole
clean_price	~~$~~3.57 ~~25% off~~
normalized_qty	~~1 gal~~ 128
store	Publix

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Data Enrichment Pipeline

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Enrich data

Field	Raw scraped data
search_term	Whole Milk
product_name	Publix Milk, Whole
clean_price	3.57
normalized_qty	128
unit_type	oz
price_per_unit	0.0278
store	Publix
category	Dairy

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Longitudinal Data

Data at a Glance

1,332
total rows

37
days: Feb 3 – Mar 11

12
basket items tracked

3
retailers compared

Basket

- Bananas, Chicken Breast, Dish Soap, Gala Apples, Ground Beef 80/20, Iceberg Lettuce, Large Eggs (12 count), Salted Butter, Spaghetti Pasta, Tomato Sauce, White Bread, Whole Milk

Publix.

ALDI

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02 EXPLORATORY ANALYSIS

Basket Cost & Item-Level Prices

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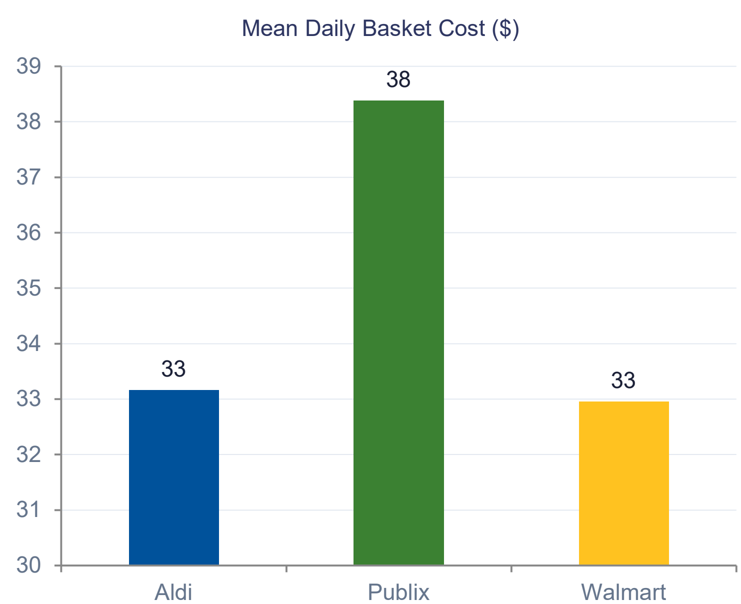
Daily Basket Cost — Summary Statistics



$33.16 / day
Aldi

$38.38 / day
Publix

$32.95 / day
Walmart

Publix runs ~$5.43/day more than Walmart and ~$5.22/day more than Aldi.

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03 STATISTICAL TESTS

RCBD, Friedman & Kruskal-Wallis

Randomized Complete Block Design (RCBD) Analysis

Study Design

- Items treated as blocks; retailers as treatment
- Removes nuisance variation between items (chicken vs bananas)
- 12 × 3 matrix with mean price per item per store across all 37 dates
- Friedman Rank Sum chosen after normality assumption was violated

Result

- Friedman chi² = 5.17, p = 0.0755
- Result: NOT significant at α = 0.05
- Post-hoc Wilcoxon: all pairs not significant

Store	Mean $/item	Mean basket/day
Aldi	$2.76	$33.16
Publix	$3.20	$38.38
Walmart	$2.75	$32.95

Why is this result not significant overall?

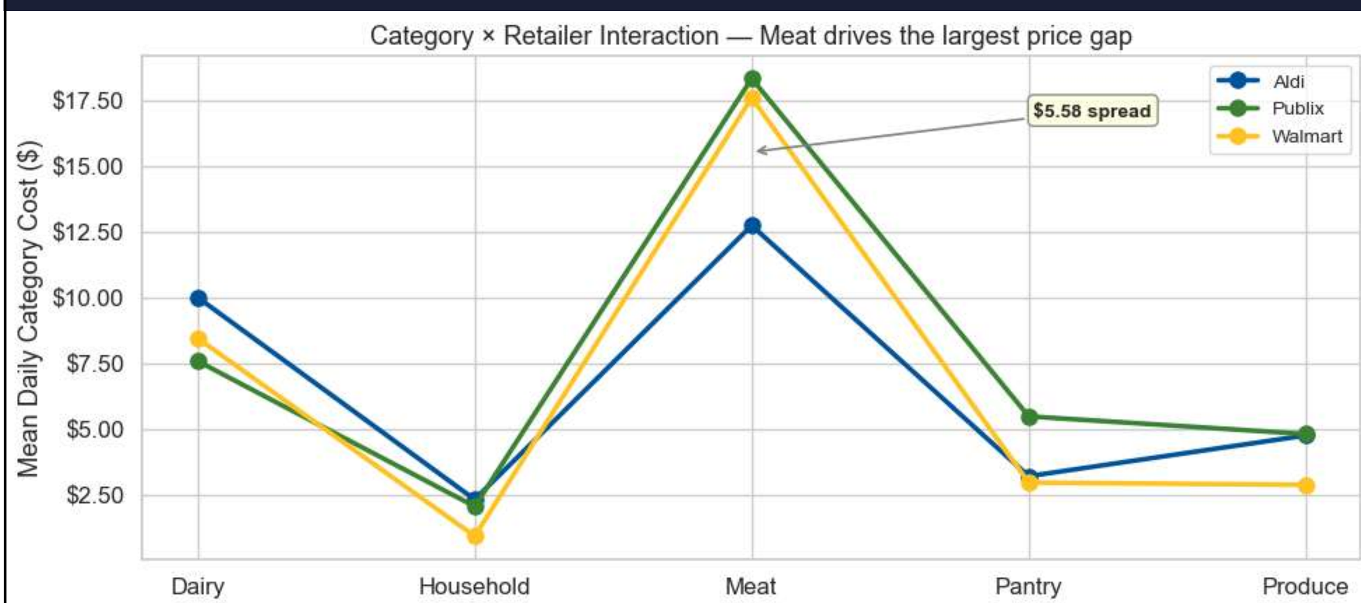
Store rankings are not consistent enough across all 12 items to reject the null globally

Does Retailer Effect Vary by Category?

Dairy	Household	Meat	Pantry	Produce
chi² = 68.49	chi² = 74	chi² = 74	chi² = 74	chi² = 56.16
$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$	$p < 0.0001$
✓ Significant	✓ Significant	✓ Significant	✓ Significant	✓ Significant

Individual categories show a significant retailer effect, i.e. store rankings ARE consistent within categories, but this result is lost when grouping all rankings together. Different retailers "win" different categories.

Does Retailer Effect Vary by Category?



04 VOLATILITY & TIMING

Price Stability & Day-of-Week Effects

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Price Volatility

Overall Coefficient of Variation (CV) by Retailer and Category

	Aldi	Publix	Walmart
Dairy	0.0154	0.0643	0.0101
Household	0.0119	0.0122	0.0261
Meat	0.0133	0.0032	0.0238
Pantry	0.0372	0.0734	0.0355
Produce	0.0962	0.0403	0.0667

- Most volatile item: Bananas
- Least volatile item: Ground Beef 80/20
- Most volatile category: Produce
- Least volatile category: Household

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Day-of-Week Effect on Basket Cost

Kruskal-Wallis Test Result

$H = 0.23$ $p = 0.9998$

NOT SIGNIFICANT ($\alpha = 0.05$)

What this means:

No day-of-week effect was detected for any store.
Per-store breakdown: Aldi p=0.97, Publix p=0.98, Walmart p=0.95.

05 OPTIMIZATION

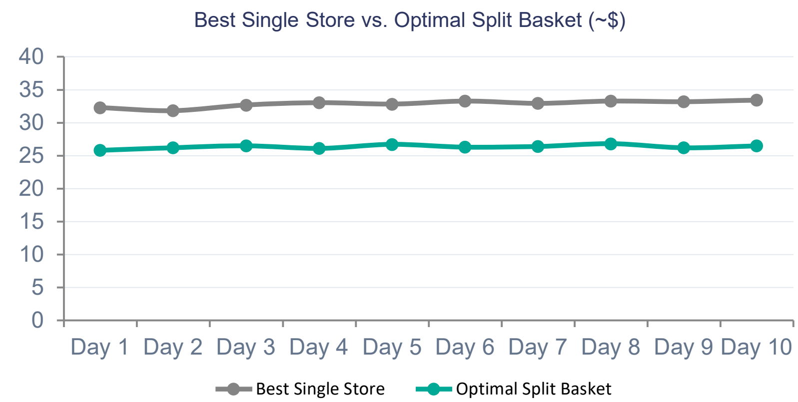
Multi-Store Split Shopping Strategy

Can Split Shopping Save Money?

$26.35
Mean optimal (split) basket cost

$6.59 / day
Avg savings vs. best single store

$7.12
Max single-day split savings



Cheapest Store by Item

- Walmart wins on Dish Soap, Gala Apples, Tomato Sauce, etc.
- Aldi wins on Bananas, Chicken, etc.
- Publix frequently wins on Salted Butter
- No store dominates more than ~6–7 items, so splitting the basket saves costs.

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Summary of Statistical Findings

	Finding	Result	Details
1	**Retailer Effect**	**NOT significant ($p = 0.076$)**	Globally, store rankings flip across items. Aldi & Walmart nearly tied on basket cost; Publix ~$5 higher.
2	**Category Interaction**	**Significant in ALL 5 categories**	Every category shows a consistent store ordering: the effect is significant, just item-dependent.
3	**Price Volatility**	**Produce most volatile; HH least**	Bananas CV = 13.6%; Ground Beef CV = 0.9%. Walmart prices fluctuate most overall.
4	**Day-of-Week**	**NOT significant ($p = 0.9998$)**	Prices do not systematically vary across the week at any store.
5	**Multi-Store Savings**	**$6.59 / day avg savings**	Cherry-picking the cheapest item per store saves ~20% vs. best single-store shopping — before accounting for travel cost.

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06 REFLECTIONS

Lessons Learned & What I'd Do Differently

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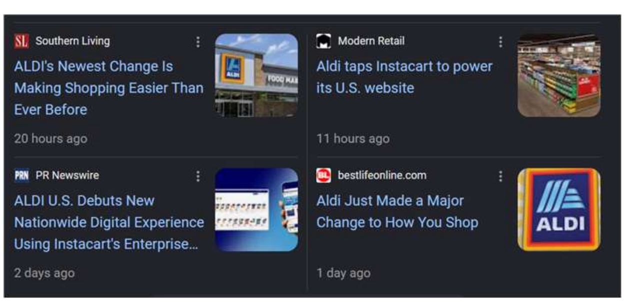
Lessons Learned & What I'd Do Differently

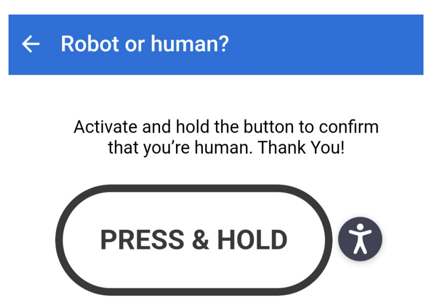
The Perils of Web Scraping

- Retailer sites change their HTML structure unpredictably. During the project, Aldi even launched an entirely new website
- Anti-bot measures (Cloudflare, rate limits) required retries and driver rotation
- Headless Chrome versioning was a recurring maintenance headache

Pivot from Zip Code Feature

- Original plan: allow per-zip-code price lookup to reflect local variation
- This worked for most of the semester
- Instacart platform changes made this impossible
- Pivoted from this feature to focus on analytics of longitudinal data





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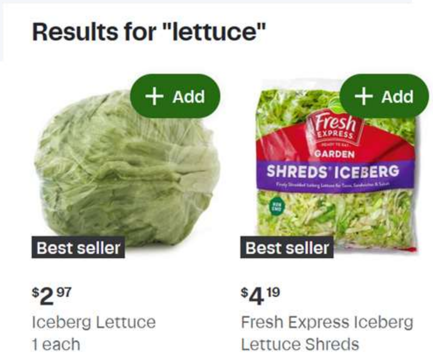
Lessons Learned & What I'd Do Differently

More Specific Search Terms

- Vague searches (e.g. 'lettuce') returned different items (head of lettuce vs. shredded lettuce bag) at different stores
- Would specify brand, size, and package type (e.g. '1 lb boneless skinless chicken breast') for cleaner unit comparisons
- Name-match fuzzy scoring helped, but a stricter allowlist of accepted product names would be better



Future: Route Cost Variable

- Split shopping assumes zero cost to visit multiple stores, which isn't true.
- Adding a 'cost of detour' variable (gas, time) would determine the true break-even savings threshold
- A $6.59/day split saving may not be worth visiting 3 stores if the round-trip adds 45 min and $3 in gas

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Questions?

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